

# Technical Documentation: AI Response Validation System

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**Project Name:** AI Response Validation System

**Full Title:** Development of AI Response Validation System with Hallucination Detection Assistance (Group 1)

**Milestone:** Milestone 4 — Complete Technical Project Document

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## 1. Project Overview

The **AI Response Quality Evaluator** is an enterprise-grade evaluation platform engineered to assess AI-generated text responses. Using **Retrieval-Augmented Generation (RAG)** evidence retrieval and a **Multi-Agent Evaluation Framework**, the platform automatically measures factual accuracy, hallucination frequency, topic relevance, and response completeness across single and batch evaluations.

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## 2. Problem Statement

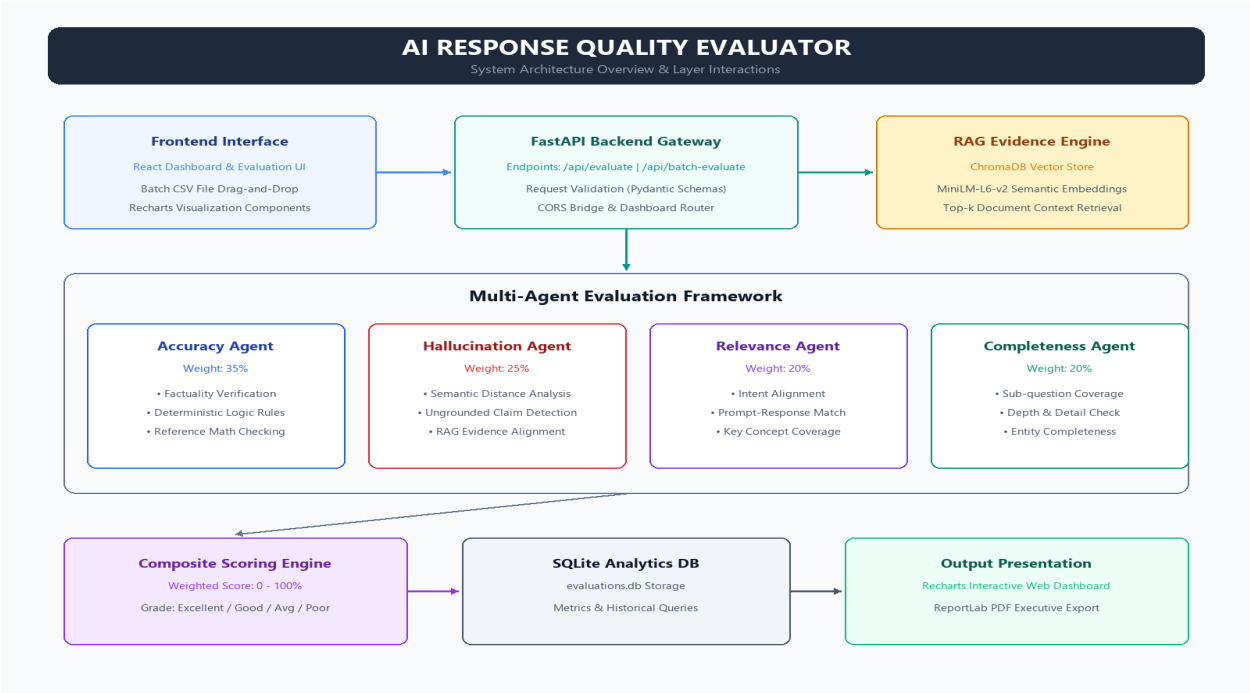
Generative AI models often produce outputs that sound plausible but contain factual errors, subtle hallucinations, or ungrounded claims. Standard n-gram metrics (like BLEU or ROUGE) fail to capture semantic correctness and factual grounding. There is a need for an automated, multi-agent evaluation pipeline that anchors LLM outputs against domain-specific knowledge bases and provides real-time quality analytics.

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## 3. Objectives

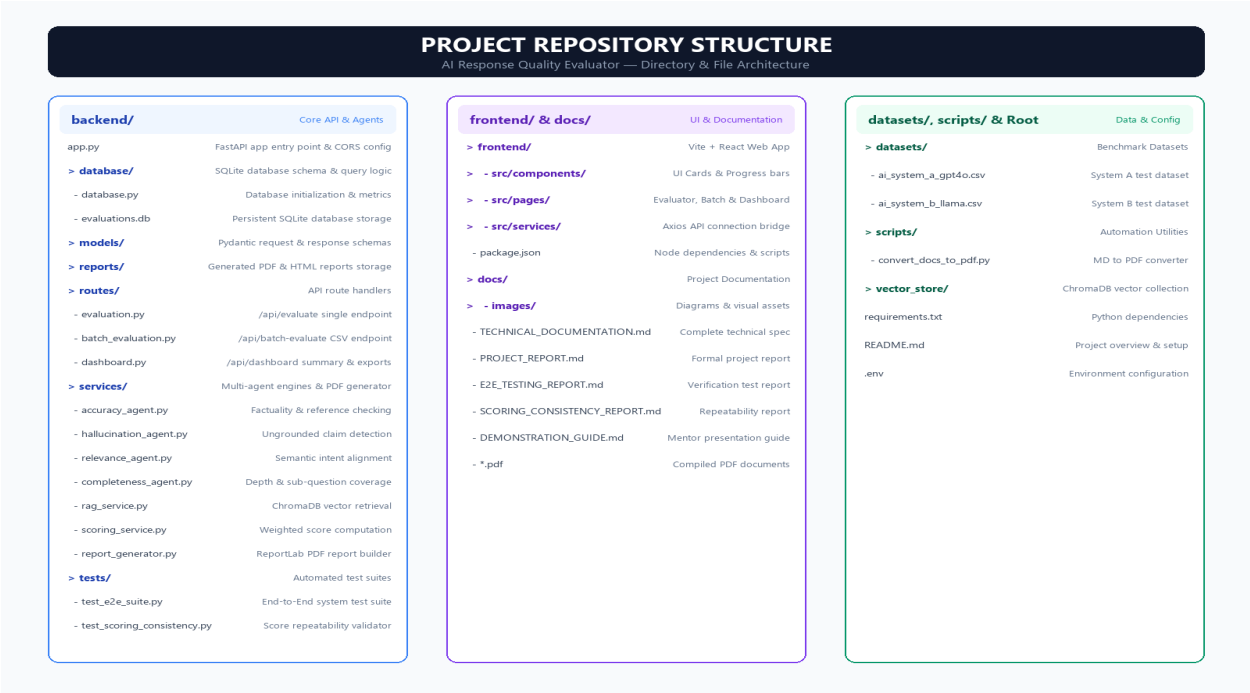
- Multi-Agent Evaluation:** Deploy specialized evaluation agents for Accuracy, Hallucination, Relevance, and Completeness.
  - RAG Context Grounding:** Integrate ChromaDB vector retrieval to verify factual claims against verified evidence.
  - Interactive Analytics Dashboard:** Visualize pass/fail distribution, metric averages, and quality trends over time with custom filters.
  - Automated PDF Export:** Support one-click generation of ReportLab executive PDF reports containing data tables, visual charts, and prompt improvement recommendations.
  - Multi-Model Benchmark Comparison:** Enable comparative evaluation of distinct AI systems (e.g., System A vs. System B).
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## 4. System Architecture



System Architecture Diagram

## 5. Folder Structure



Project Folder Structure Graphic

## 6. Installation Guide

### 6.1 Prerequisites

- \* Python 3.10+
- \* Node.js 18+ & npm
- \* SQLite3

## 6.2 Backend Setup

```
# 1. Activate virtual environment venv\Scripts\activate # Windows # 2. Install dependencies
pip install -r requirements.txt # 3. Start FastAPI server uvicorn backend.app:app --reload
--port 8000
```

\*Backend API available at: <http://localhost:8000> (API Docs at <http://localhost:8000/docs>).\*

## 6.3 Frontend Setup

```
# 1. Navigate to frontend folder cd frontend # 2. Install dependencies npm install # 3.
Launch dev server npm run dev
```

\*Frontend app available at: <http://localhost:5173>.\*

# 7. API Documentation

### 7.1 Single Evaluation Route

\* **Endpoint:** [POST /api/evaluate](#)

\* **Request Body:**

```
{ "question": "What causes malaria?", "response": "Malaria is caused by Plasmodium parasites
transmitted by Anopheles mosquitoes.", "model": "GPT-4o", "dataset": "Single Query" }
```

\* **Response:** Returns 4-agent score objects, RAG evidence, composite score, grade, and reasoning.

### 7.2 Batch Evaluation Route

\* **Endpoint:** [POST /api/batch-evaluate](#)

\* **Payload:** [multipart/form-data](#) with CSV file ([file](#)).

\* **Response:** Summary stats ([total\\_records](#), [average\\_score](#)) and array of itemized evaluation objects.

### 7.3 Dashboard Analytics Routes

\* [GET /api/dashboard/summary](#) — Aggregate summary stats and metric averages.

\* [GET /api/dashboard/evaluations](#) — Query evaluation records with filter criteria.

\* [GET /api/dashboard/trends](#) — Daily/weekly quality trend data over time.

\* [GET /api/dashboard/report/pdf](#) — Stream downloadable ReportLab PDF executive report.

# 8. Database Schema

Table Name: [evaluations](#) (SQLite)

| <a href="#">id</a>                  | INTEGER | PRIMARY KEY<br>AUTOINCREMENT | Unique record identifier     |
|-------------------------------------|---------|------------------------------|------------------------------|
| <a href="#">question</a>            | TEXT    | NOT NULL                     | User question / prompt       |
| <a href="#">response</a>            | TEXT    | NOT NULL                     | AI model response            |
| <a href="#">accuracy_score</a>      | REAL    | DEFAULT 0                    | Accuracy metric (0–100)      |
| <a href="#">relevance_score</a>     | REAL    | DEFAULT 0                    | Relevance metric (0–100)     |
| <a href="#">hallucination_score</a> | REAL    | DEFAULT 0                    | Hallucination metric (0–100) |

|                        |         |                  |                                                                                                               |
|------------------------|---------|------------------|---------------------------------------------------------------------------------------------------------------|
| completeness_score     | REAL    | DEFAULT 0        | Completeness metric (0–100)                                                                                   |
| final_score            | REAL    | DEFAULT 0        | Composite score (0–100)                                                                                       |
| grade                  | TEXT    |                  | Verdict ( <a href="#">Excellent</a> , <a href="#">Good</a> , <a href="#">Average</a> , <a href="#">Poor</a> ) |
| hallucination_detected | INTEGER | DEFAULT 0        | Flag (1 if hallucination detected, 0 otherwise)                                                               |
| timestamp              | TEXT    |                  | ISO Timestamp                                                                                                 |
| evaluation_mode        | TEXT    | DEFAULT 'single' | Mode ( <a href="#">single</a> or <a href="#">batch</a> )                                                      |
| model                  | TEXT    |                  | Model identifier (e.g. <a href="#">GPT-4o</a> , <a href="#">Llama-3</a> )                                     |
| dataset                | TEXT    |                  | Dataset origin name                                                                                           |

## 9. RAG Workflow

- Query Ingestion:** User question is received by `rag_service.py`.
- Vector Retrieval:** Embedding vector is generated using `sentence-transformers/all-MiniLM-L6-v2` and passed to **ChromaDB**.
- Context Assembly:** Top- $k$  nearest context passages are retrieved.
- Agent Evidence Injection:** Retrieved context passages are formatted into structured reference evidence for the evaluation agents.

## 10. Agent Descriptions

- Accuracy Agent** (`[accuracy_agent.py](file:///c:/Users/hi/OneDrive/Desktop/AI_Response_Quality_Evaluator/backend/services/accuracy_agent.py)`):  
\* Evaluates factual correctness using LLM reasoning and deterministic math/logic rules.
- Hallucination Agent** (`[hallucination_agent.py](file:///c:/Users/hi/OneDrive/Desktop/AI_Response_Quality_Evaluator/backend/services/hallucination_agent.py)`):  
\* Computes semantic distance between response claims and retrieved RAG context. Flagged if ungrounded claims are detected.
- Relevance Agent** (`[relevance_agent.py](file:///c:/Users/hi/OneDrive/Desktop/AI_Response_Quality_Evaluator/backend/services/relevance_agent.py)`):  
\* Evaluates semantic similarity, keyword coverage, and intent alignment between prompt and response.
- Completeness Agent** (`[completeness_agent.py](file:///c:/Users/hi/OneDrive/Desktop/AI_Response_Quality_Evaluator/backend/services/completeness_agent.py)`):  
\* Checks coverage of key sub-questions, entities, and required details.

## 11. Dashboard Explanation

The **Analytics Dashboard** (`[Dashboard.jsx](file:///c:/Users/hi/OneDrive/Desktop/AI_Response_Quality_Evaluator/frontend/src/pages/Dashboard.jsx)`) provides a real-time command center:

- \* **Metrics Cards:** Total Evaluations, Pass Rate, Average Score, Hallucination Rate.

- \* **Result Distribution Pie Chart:** Visual breakdown of Excellent, Good, Average, and Poor grades.
  - \* **Dimension Bar Chart:** Comparative visual of Accuracy vs Relevance vs Completeness vs Hallucination.
  - \* **Quality Trend Line Chart:** Historical trend analysis across dates.
  - \* **Filter Controls:** Dynamic filtering by Model, Dataset, Evaluation Mode, and Date Range.
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## 12. Report Generation

The report engine ([[report\\_generator.py](#)](file:///c:/Users/hi/OneDrive/Desktop/AI\_Response\_Quality\_Evaluator/backend/services/report\_generator.py)) uses **ReportLab** to build publication-grade PDF documents containing:

- \* Project details & metadata header.
  - \* Summary statistical tables.
  - \* Programmatically rendered Pie and Bar chart graphics.
  - \* Flagged hallucination warnings table.
  - \* Itemized individual evaluation records.
  - \* Actionable prompt engineering & model fine-tuning recommendations.
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## 13. Testing Results

- \* **End-to-End Test Suite** ([[test\\_e2e\\_suite.py](#)](file:///c:/Users/hi/OneDrive/Desktop/AI\_Response\_Quality\_Evaluator/backend/tests/test\_e2e\_suite.py)): **7/7 PASSED** (100% test coverage across RAG, agents, DB, PDF export, error handling).
  - \* **Scoring Consistency** ([[test\\_scoring\\_consistency.py](#)](file:///c:/Users/hi/OneDrive/Desktop/AI\_Response\_Quality\_Evaluator/backend/tests/test\_scoring\_consistency.py)): Evaluated 3 repeated iterations. Result:  $\sigma = 0.0000$  and  $CV = 0.00\%$  (100% deterministic score repeatability).
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## 14. Limitations

- Domain Context Dependency:** Factual accuracy verification relies on the quality and scope of documents indexed in the ChromaDB vector store.
  - Third-Party LLM Rate Limits:** Complex non-math prompts rely on external LLM APIs (OpenAI/Anthropic) subject to rate limits and API quota availability.
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## 15. Future Work

- LLM-as-a-Judge Provider Switching:** Enable seamless switching between OpenAI, Anthropic Claude, and local Ollama models.
- WebSocket Real-time Streaming:** Stream real-time progress updates for multi-thousand row batch evaluation runs.
- Custom Weighting & Threshold Management UI:** Allow users to adjust evaluation dimension weights dynamically from the Dashboard.